

Does Double/Debiased Machine Learning Change the AJR Estimate?

A Coefficient-Comparison Test in a Small-Sample IV Design

Wenjing Gao¹ and Edith Jiang¹

¹EC610I – Machine Learning in Economics, Wilfrid Laurier University

July 2026

Abstract

We ask a narrow, directly testable question: in Acemoglu, Johnson, and Robinson’s (2001, AJR) small-sample instrumental-variables design ($N = 64$ former colonies), does replacing the linear first-stage and outcome models with double/debiased machine learning (DML; Chernozhukov et al., 2018) produce a coefficient that is *statistically different* from the conventional 2SLS estimate? We estimate the institutional effect on log GDP per capita under three nested control specifications, each with 2SLS and five DML nuisance learners (Lasso, Ridge, Elastic Net, Random Forest, Gradient Boosting), and report an approximate difference-in-coefficients test between every DML estimate and its 2SLS counterpart from the identical specification and sample. Across all 15 comparisons, no DML coefficient is statistically distinguishable from 2SLS at the 5% level. However, the three regularized linear learners produce similar coefficients with standard errors 25–45% smaller than 2SLS. Tree-ensemble learners move the point estimate substantially (in one specification, more than doubling it) but their own standard errors inflate so much that even this large a shift fails to reach significance – evidence of estimator instability, not a competing causal finding. These results suggest that regularized linear DML can improve estimated precision without changing the underlying causal conclusion, whereas tree-based nuisance learners are unstable in this small sample.

1 Introduction

AJR (2001) instrument modern institutional quality with log settler mortality to identify the causal effect of institutions on long-run GDP, using 64 former European colonies. Their 2SLS estimate – 0.93 in the corrected 2012 data – is a canonical result in development economics. Rather than asking the broad question of whether “machine learning improves causal inference” in the abstract, we ask a single, sharply defined and directly testable question:

If we replace the linear nuisance functions inside AJR’s IV estimator with machine learning, does the resulting coefficient differ, in a formal statistical sense, from the original 2SLS estimate?

This reframing matters because *whether the coefficient differs* and *how precisely it is estimated* are two separate questions. A large difference in point estimates is not, by itself, informative if both estimates carry wide uncertainty; a small difference in point estimates combined with a smaller standard error is informative even when the two coefficients are not formally distinguishable. Our contribution is to make this comparison explicit: for every machine-learning estimator we consider, we report the point estimate, an approximate equality test against 2SLS, and the percentage change in standard error. This lets us describe, coefficient by coefficient,

whether – and in what sense – DML changes on the original 2SLS approach.

Our focus on coefficient comparison distinguishes this paper from studies that evaluate machine learning primarily by predictive accuracy. In causal applications the object of interest is the structural parameter β_1 , not the fitted outcome, so predictive performance alone is not sufficient evidence that a learner has improved the analysis: a nuisance learner can fit y , R , or M well and still produce a badly behaved $\hat{\beta}_1$ if it introduces regularization bias into the residualization step. Accordingly, our evaluation throughout emphasizes statistical equality of the coefficient and changes in estimation precision, rather than improvements in out-of-sample prediction.

2 Data and Baseline

Our sample is 64 former European colonies from the AJR (2012) reply dataset. The endogenous regressor is protection against expropriation risk (ICRG, 1985–1995, 0–10 scale). The instrument is log settler mortality. The outcome is log GDP per capita.

We reproduce AJR (2012) Table 1B column 1 exactly: $\hat{\beta}_1 = 0.929$, conventional SE = 0.156, HC1-robust SE = 0.173, first-stage $F = 23.3$. This no-controls estimate serves as a reference point throughout the paper – a benchmark estimate against which the fragility in-

roduced by adding controls can be judged – though our formal equality tests (Section 4) compare DML directly to the *controlled* 2SLS estimate from the same specification, not to this no-controls benchmark.

We estimate the causal effect under three nested control specifications that span AJR’s own robustness checks:

- **Spec 1 (latitude only, $N = 64$):** the one control AJR themselves treat as unambiguously pre-instrument.
- **Spec 2 (AJR robustness controls, $N = 62$):** latitude, malaria prevalence (1994), and % European descent (1975) – AJR’s own robustness-check controls.
- **Spec 3 (AJR + mediators, $N = 56$):** Spec 2 plus log population density, log trade, FDI, and life expectancy.

Specification 3 is best read as a sensitivity exercise rather than a like-for-like extension of Specifications 1–2: because the added variables may themselves be channels through which institutions affect GDP, conditioning on them can shift the estimand toward a more conditional institutional effect rather than AJR’s total effect. Within Specification 3, the comparison between DML and 2SLS is still valid (both estimators target the same conditional parameter), but that parameter should not be read as directly comparable to the total effect recovered in Specifications 1–2. Each specification uses its own complete-case sample, which is why N varies across specifications.

3 Method

3.1 DML in the IV Setting

For each specification we estimate the causal effect two ways: we estimate the original AJR specification using 2SLS and report HC1-robust standard errors, and we apply DML in the partially linear IV model of Chernozhukov et al. (2018). Writing y_i for log GDP, R_i for institutional quality, M_i for log settler mortality, and X_i for the control vector:

$$y_i = \beta_1 R_i + g_y(X_i) + \varepsilon_i, \quad (1)$$

$$R_i = m_R(X_i) + \xi_i, \quad M_i = m_M(X_i) + \eta_i. \quad (2)$$

Each nuisance function (g_y , m_R , m_M) is estimated by the same machine-learning method on a training fold and residualized on a held-out fold, using $K = 5$ -fold cross-fitting. The coefficient is recovered from the cross-fitted residuals,

$$\hat{\beta}_1 = \frac{\sum_i \tilde{M}_i \tilde{y}_i}{\sum_i \tilde{M}_i \tilde{R}_i}, \quad (3)$$

Intuitively, DML first removes the variation explained by the control variables from the outcome, the endogenous regressor, and the instrument, using the fitted nuisance functions; the causal coefficient is then estimated using only the residual variation that remains after this residualization step. Cross-fitting ensures that the same observations are never used both to train the nuisance models and to form the residuals used in estimation, which limits the overfitting bias that would otherwise result from a flexible learner fitting noise in the training fold and passing that noise into $\hat{\beta}_1$, with a heteroskedasticity-robust orthogonal-score standard error built from the Neyman-orthogonal moment $\psi_i(\beta) = \tilde{M}_i(\tilde{y}_i - \beta \tilde{R}_i)$. We compare this DML standard error against 2SLS’s HC1-robust standard error; both are heteroskedasticity-robust, but they are constructed differently (one from a sandwich formula on cross-fitted residuals, the other from the classical HC1 correction), so the comparison is between two robust standard errors of different provenance, not a like-for-like recomputation of the same object. We use five nuisance learners: Lasso, Ridge, and Elastic Net (regularization chosen by inner cross-validation), Random Forest (300 trees, depth 4), and Gradient Boosting (150 trees, depth 3).

3.2 The Coefficient-Comparison Test

For every DML estimate we compute an approximate difference-in-coefficients statistic against the 2SLS estimate from the identical specification and sample:

$$z = \frac{\hat{\beta}_1^{\text{DML}} - \hat{\beta}_1^{\text{2SLS}}}{\sqrt{(\text{SE}^{\text{DML}})^2 + (\text{SE}^{\text{2SLS}})^2}}. \quad (4)$$

This statistic is analogous in spirit to a Hausman-style comparison, but it is not a formal Hausman test: it treats the two estimators’ sampling distributions as independent and does not estimate the covariance between them, since 2SLS and DML are computed on the same underlying sample and a tractable joint covariance is not available. This is a simplifying approximation, not necessarily a conservative one – ignoring a nonzero covariance can move the test in either direction depending on its sign. We report the resulting two-sided p -value as our primary evidence on coefficient equality. Confidence-interval overlap is reported alongside as a visual diagnostic, not as an independent formal test; throughout the paper, the formal comparison is the p -value from Equation 4.

Estimator	$\hat{\beta}_1$	SE	SE Δ	p
<i>Spec 1: Latitude only (N=64, F=13.5)</i>				
2SLS	0.962	0.232	—	—
DML Lasso	0.956	0.228	−1.6%	0.99
DML Ridge	0.948	0.229	−1.4%	0.96
DML Elastic Net	0.959	0.230	−1.1%	0.99
DML Random Forest	0.910	0.247	+6.5%	0.88
DML Gradient Boost.	0.921	0.294	+26.5%	0.91
<i>Spec 2: AJR robustness controls (N=62, F=3.3)</i>				
2SLS	0.660	0.368	—	—
DML Lasso	0.559	0.257	−30.2%	0.82
DML Ridge	0.609	0.275	−25.1%	0.91
DML Elastic Net	0.572	0.268	−27.2%	0.85
DML Random Forest	1.366	0.896	+143.5%	0.47
DML Gradient Boost.	1.073	0.695	+88.9%	0.60
<i>Spec 3: AJR + mediators (N=56, F=2.8)</i>				
2SLS	0.450	0.336	—	—
DML Lasso	0.422	0.236	−29.6%	0.95
DML Ridge	0.368	0.184	−45.1%	0.83
DML Elastic Net	0.413	0.227	−32.4%	0.93
DML Random Forest	0.694	0.341	+1.6%	0.61
DML Gradient Boost.	0.610	0.313	−6.8%	0.73

Table 1: Coefficient comparison and approximate equality tests, DML vs. 2SLS. SE is HC1-robust for 2SLS and orthogonal-score for DML. p is the two-sided p -value from Equation 4. Every one of the 15 DML-vs-2SLS pairs also has overlapping 95% confidence intervals (a secondary, visual check; see Figure 1). Bold SE changes mark the $\geq 25\%$ reductions achieved by all three regularized linear learners in Specs 2–3.

4 Results

4.1 Main Table: Coefficients and Coefficient-Comparison Tests

Table 1 is the central empirical result of the paper. For every specification and every DML learner it reports the point estimate, the standard error, the percentage change in SE relative to 2SLS, and the p -value from Equation 4.

No coefficient in Table 1 is statistically different from its same-specification 2SLS counterpart: all 15 p -values exceed 0.46. This means DML does not produce a formally different causal conclusion than 2SLS at any specification we consider.

Where the two approaches diverge is precision, not the coefficient itself. In Specifications 2 and 3 – where the 2SLS first stage is weakest ($F = 3.3$ and 2.8 , both below the conventional threshold of 10) – the three regularized linear DML learners produce coefficients statistically indistinguishable from 2SLS while cutting the standard error by 25% to 45%. In Specification 1, where the instrument is already strong ($F = 13.5$), the SE changes are much smaller (−1% to −2%).

Tree-ensemble learners move the point estimate substantially – in Specification 2, Random Forest’s coef-

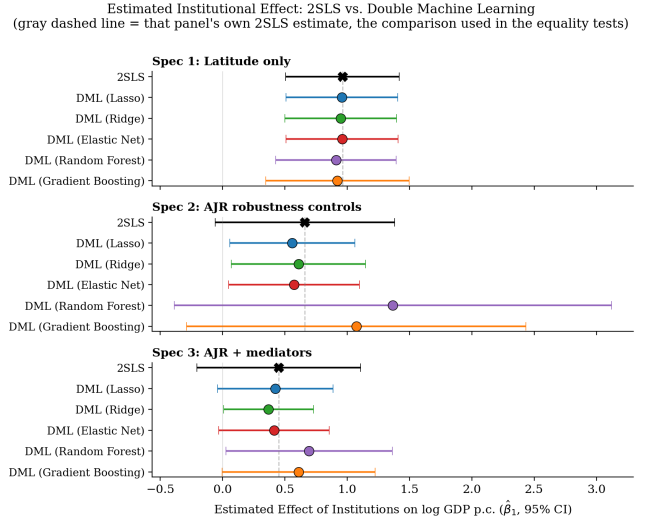


Figure 1: Estimated institutional effect: 2SLS and DML coefficients across specifications. Points show coefficient estimates and horizontal bars show 95% confidence intervals. The black $\times$ is the same-specification 2SLS estimate. The dashed line in each panel marks that panel’s own 2SLS coefficient, the benchmark used in the equality test of Table 1.

ficient (1.366) is more than double the 2SLS estimate (0.660) – but the coefficient remains statistically indistinguishable from 2SLS ($p = 0.47$), because Random Forest’s own standard error (0.896) is more than double 2SLS’s (0.368). A large point-estimate shift combined with a large standard error is not statistical evidence of a different effect.

4.2 Main Figure

Figure 1 plots every estimator’s point estimate and 95% confidence interval, grouped by specification, with 2SLS shown as a black $\times$ in each panel. The pattern in Table 1 is directly visible: the three linear DML learners sit close to the 2SLS estimate with visibly tighter intervals, while the two tree learners are noticeably shifted with much wider intervals that still comfortably contain the 2SLS point estimate.

4.3 Interpretation Across Specifications

Read together, the three specifications reveal a consistent pattern rather than three unrelated snapshots. When only latitude is controlled for (Specification 1), the first-stage relationship remains relatively strong ($F = 13.5$), and DML produces coefficients and standard errors nearly identical to 2SLS – there is little for a more flexible nuisance learner to improve upon. As additional controls are introduced in Specification 2, the classical first stage weakens considerably ($F = 3.3$), and this is precisely where the regularized linear learners begin to produce noticeably smaller standard errors while the

coefficient itself remains statistically indistinguishable from 2SLS. Specification 3 introduces variables that may lie on the causal pathway from institutions to income, so both 2SLS and DML there are best read as estimating a more conditional effect (Section 2). Although the point estimate declines further in this specification, the comparison between DML and 2SLS *within* it is unchanged from the pattern in Specification 2: no learner produces a statistically different coefficient, and the linear learners continue to deliver the same coefficient more precisely. The precision advantage of linear DML, in other words, tracks the weakness of the classical first stage across specifications, not the specific set of controls used.

5 Discussion

5.1 Linear Learners: Similar Coefficients, Smaller Standard Errors

The standard-error reduction achieved by Lasso, Ridge, and Elastic Net appears consistently across two independent specifications (2 and 3), is largest exactly where the classical first stage is weakest, and holds while the coefficient itself barely moves (differences of 0.03–0.09 in Specification 2, well within one standard error of either estimator). The pattern is consistent with regularization stabilizing nuisance-function estimation when controls are moderately collinear with the instrument, as AJR’s robustness controls are – though we do not separately test this mechanism against alternatives, and it should be read as a plausible interpretation rather than an identified mechanism. In Specification 1, where the classical first stage is already strong, the linear DML learners offer almost no precision advantage. This is economically intuitive rather than merely descriptive: with a single control, the nuisance functions relating latitude to the outcome, the treatment, and the instrument are close to linear and easy to estimate with ordinary least squares. Replacing that linear regression with a regularized machine-learning method provides little additional flexibility to exploit, so DML has little room to improve on 2SLS. The precision benefit becomes visible only once additional, more collinear controls are introduced, where regularization can reduce estimation noise in the nuisance functions while leaving the recovered coefficient largely unchanged.

5.2 Tree Learners: Large Shifts Without Statistical Support

Random Forest’s near-doubling of the coefficient in Specification 2 could be read, on its face, as evidence that a more flexible model captures something 2SLS misses. The coefficient-comparison test cautions against that reading: the difference is not statistically supported

($p = 0.47$), because Random Forest’s own standard error (0.896) is more than double 2SLS’s (0.368). At $N \approx 60$, a several-hundred-tree ensemble has few effective observations per leaf, and we verified this instability is not an artifact of our specific hyperparameters – alternative depth and leaf-size specifications produce comparably large standard errors. The large tree-based point-estimate shifts are better interpreted as small-sample estimator instability than as evidence of a different causal effect.

The tree learners illustrate a broader lesson that is easy to overlook when evaluating machine learning by point estimates alone: a more flexible learner is not necessarily a better learner in a causal setting. Random Forest and Gradient Boosting produce substantially different point estimates from 2SLS, but those differences are accompanied by much larger standard errors – in other words, the additional flexibility increases variance faster than it reduces any bias from nuisance-function misspecification. For a sample of roughly sixty observations, the data simply do not contain enough information to support a highly adaptive nuisance model with hundreds of trees; the classical bias-variance tradeoff resolves here in favor of the simpler, more constrained linear learners.

5.3 In What Sense Does DML Improve on 2SLS?

Two distinct answers, by learner class. For the regularized linear learners, DML produces similar coefficients with materially smaller estimated standard errors than 2SLS, in the specifications where 2SLS is weakest. For the tree-ensemble learners, neither the coefficient nor the precision improves: the point estimates move substantially, but the accompanying standard errors are too large to treat those movements as informative. Whether DML “improves on” 2SLS in this application therefore depends entirely on which nuisance-learner class is used, and answering that question requires the coefficient-by-coefficient comparison reported here, rather than relying solely on visual differences in point estimates.

6 Limitations

Three limitations deserve emphasis. First, our coefficient-comparison test (Equation 4) does not estimate the covariance between the 2SLS and DML estimators, even though both are computed on the same underlying sample; a paired bootstrap that resamples both estimators jointly would give a more exact test, and we do not implement that here. Second, at $N = 56$ –64, both the DML sandwich standard error and our normal-approximation test rely on asymptotic justifications that warrant some caution at this sample size. The sample contains only 56–64 observations after complete-case deletion, and although DML remains theoretically valid under the

regularity conditions of Chernozhukov et al. (2018), its finite-sample behavior depends heavily on the quality of the underlying nuisance-function estimates – which is, in fact, the central empirical question this paper investigates. Our findings should therefore be read as evidence from a particularly challenging small-sample application, not as a general statement about DML’s performance at larger sample sizes. Third, the first stage itself is weak in Specifications 2 and 3 (F below the conventional threshold of 10), so point estimates from either estimator in those specifications should be interpreted cautiously, independent of which method produced them.

7 Conclusion

Across three specifications and five machine-learning nuisance learners, no DML coefficient is statistically distinguishable from the corresponding 2SLS estimate. DML does not change the substantive conclusion of AJR’s analysis. What differs by learner class is precision: regularized linear DML learners produce similar coefficients with materially smaller estimated standard errors than 2SLS, in the specifications where the classical first stage is weakest. Tree-ensemble learners add instability rather than value – their point estimates shift substantially, but the accompanying standard errors are too large for those shifts to carry statistical weight.

More broadly, our results suggest that evaluating machine learning in economics should not focus only on whether coefficients become larger or smaller. Instead, researchers should ask whether the estimated causal effect changes in a statistically meaningful way, and whether the uncertainty surrounding that estimate improves. In the AJR application, machine learning does not overturn the original economic conclusion, but regularized linear DML suggests that a similar causal estimate can sometimes be obtained more precisely – which is a narrower, but more defensible, claim than saying machine learning “improves” causal inference in general.

References

- [1] Acemoglu, D., Johnson, S., Robinson, J.A. (2001). The Colonial Origins of Comparative Development. *AER*, 91(5), 1369–1401.
- [2] Acemoglu, D., Johnson, S., Robinson, J.A. (2012). The Colonial Origins of Comparative Development: Reply. *AER*, 102(6), 3077–3110.
- [3] Chernozhukov, V., et al. (2018). Double/Debiased Machine Learning for Treatment and Structural Parameters. *Econometrica J.*, 21(1), C1–C68.

- [4] Curtin, P.D. (1989). *Death by Migration*. Cambridge University Press.
- [5] Hausman, J.A. (1978). Specification Tests in Econometrics. *Econometrica*, 46(6), 1251–1271.

Appendix: Standard Errors by Specification and Estimator

Figure 2 shows the same standard errors reported in Table 1, organized as a bar chart grouped by specification and estimator, to make the SE pattern across the 18 estimates easier to scan visually alongside the coefficient-level comparison in Figure 1.

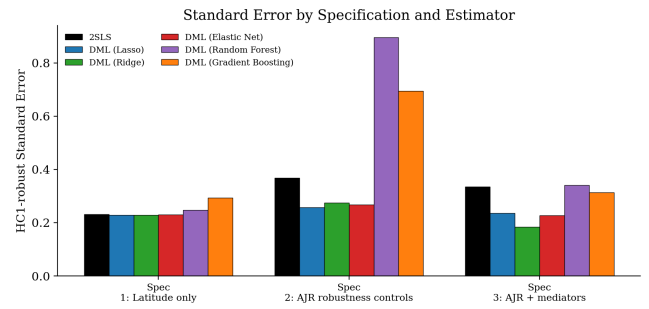


Figure 2: HC1-robust (2SLS) and orthogonal-score (DML) standard errors by specification and estimator. Bars correspond to the SE column of Table 1.